

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | Kent, OH (open to NYC relocation)

PROFESSIONAL SUMMARY

Applied AI engineer (2.5+ years) with production RAG and multi-agent LLM systems in healthcare - bringing full-stack delivery discipline and 0-to-1 builder experience to Glean's founding FDE cohort. Master's degree (Kent State University). Track record of shipping complete AI systems independently: ~35% retrieval precision gain in healthcare RAG, multi-agent healthcare claims pipeline with schema-validated agent contracts, and two independently shipped products (Manga Lens Chrome extension with 29-site coverage, E-Farming marketplace with 80-120 active users). Experienced in identifying business problems and building AI-powered solutions end-to-end - from data to evaluation to deployment - with stakeholder-facing communication in compliance-sensitive environments.

CORE COMPETENCIES

LLM Application Development • Retrieval-Augmented Generation (RAG) • Multi-Agent Systems • Agentic Workflows • 0-to-1 Product Delivery • FastAPI / Flask • Docker • Python • TypeScript / React • Evaluation Pipelines • Enterprise AI Communication • End-to-End Delivery

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare Technology, AI Consulting, SaaS Healthcare | USA

- Built **Retrieval-Augmented Generation (RAG)** systems for clinical knowledge retrieval and healthcare documentation search; implemented recursive semantic chunking and transformer-based embeddings; improved contextual retrieval precision by ~35% and reduced irrelevant retrieval by >30%. Retrieval-grounded response alignment exceeded 90% in evaluation.
- Developed **agentic LLM workflows** for multi-step healthcare queries with structured reasoning, tool discipline, and grounding rules; improved agent response stability by ~25%.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall by 15-20% on high-risk categories while holding precision >90%.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**; reduced post-deployment defects by ~30%.
- Maintained HIPAA-conscious data governance: de-identification, data lineage documentation, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Oil & Gas, Enterprise ERP | Hyderabad, India

- Optimized SQL and **T-SQL stored procedures** for the Synthesis Order-to-Cash platform; reduced query execution time ~20% and data retrieval latency ~25%.
- Built **CI/CD pipelines with Jenkins** for schema updates and deployments; reduced errors >30% and improved release cycle time ~35-40%.
- Designed performance dashboards using **SQL Server DMVs and Grafana**; cut incident recurrence ~25%.

Suvidha Foundation

Jan 2022 - Mar 2022

Machine Learning Engineer - Nonprofit Tech, Video Analytics | Hyderabad, India

- Built a **transformer-based video summarization** system across 5,000+ sessions; reduced manual review time by 60-70% (hours → <5 minutes per video).
- Implemented **document Q&A with RAG**; reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed the stack as a **Flask-based API** with lightweight interface for non-technical staff - full end-to-end ownership.

SELECTED PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization → Validation → Consistency Analysis → Duplicate Detection → Fraud Risk Scoring) with **schema-validated JSON contracts between agents** to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; approximate-nearest-neighbor similarity search for duplicate detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped · Chrome Web Store

Full-stack browser extension built independently - Manifest V3, content scripts, service workers. Integrates **four AI vision providers** (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama). Multi-section capture pipeline, seven-day translation cache, per-domain configs for 29 sites. Complete 0-to-1 product shipped and maintained solo.

E-Farming Digital Marketplace Founder & Full-Stack Developer

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded **80-120 active users** in phase one; cart, reviews, and community blogging features to drive repeat engagement. End-to-end founder ownership from design to deployment.

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app coordinating multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for visuals). Introduced **intermediate structured prompt transformation layers** - improved contextual alignment ~**30%** and first-pass image success rate ~**25-30%**.

EDUCATION

Master's Degree - *Kent State University, Ohio, USA*

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, multi-agent systems, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, observability/logging

Work Authorization: F-1 OPT - authorized to work in the United States now; future sponsorship required